

P8107 FINAL EXAM

Fall 2024

Name: <Write your name here>

UNI: <Write your uni here>

For this exam you allowed up to four sheets of paper (front and back). No other reference materials may be used. You have up to 3 hours to complete this exam. Be sure to show all your work. You may attach additional pages to your exam paper if needed.

Honor code:

I have not and will not give or receive aid in this examination nor have I concealed any violation of the University Honor Code.

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1. (20 points) Random variables $X_1, X_2, \dots, X_n$ are iid $N(\mu, \sigma^2)$ and, independently of the X 's, $Y_1, Y_2, \dots, Y_n$ are iid $N(\mu, 2\sigma^2)$. Assuming that μ is a known constant, the MLE of σ^2 is

$$\hat{\sigma}^2 = \frac{1}{2n} \sum_{i=1}^n (X_i - \mu)^2 + \frac{1}{4n} \sum_{i=1}^n (Y_i - \mu)^2.$$

- Calculate the mean and variance of $\hat{\sigma}^2$.
- Is $\hat{\sigma}^2$ unbiased for σ^2 ? Is it consistent? Justify your answers.

(a) we know that $\frac{X_i - \mu}{\sigma} \sim N(0,1)$ so $\left(\frac{X_i - \mu}{\sigma}\right)^2 \sim \chi_1^2$ and so $\sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma}\right)^2 \sim \chi_n^2$

By properties of the χ^2 dist, we know $E\left[\sum \frac{(X_i - \mu)^2}{\sigma^2}\right] = n$ Also $\text{Var}\left(\sum \frac{(X_i - \mu)^2}{\sigma^2}\right) = 2n$

$$E\left[\frac{1}{2n} \sum \frac{(X_i - \mu)^2}{\sigma^2}\right] = \frac{1}{2} \quad \text{Var}\left(\frac{1}{2n} \sum \frac{(X_i - \mu)^2}{\sigma^2}\right) = \frac{1}{2n}$$

$$\text{So } E\left[\frac{1}{2n} \sum (X_i - \mu)^2\right] = \frac{\sigma^2}{2}$$

$$\text{Var}\left(\frac{1}{2n} \sum (X_i - \mu)^2\right) = \frac{\sigma^4}{2n}$$

Similarly, $\frac{Y_i - \mu}{\sqrt{2}\sigma} \sim N(0,1)$ $\frac{(Y_i - \mu)^2}{2\sigma^2} \sim \chi_1^2$ so $\sum_{i=1}^n \frac{(Y_i - \mu)^2}{2\sigma^2} \sim \chi_n^2$

$$\text{So } E\left[\sum \frac{(Y_i - \mu)^2}{2\sigma^2}\right] = n \quad \text{and} \quad \text{Var}\left(\sum \frac{(Y_i - \mu)^2}{2\sigma^2}\right) = 2n$$

$$E\left[\frac{1}{4n} \sum \frac{(Y_i - \mu)^2}{\sigma^2}\right] = \frac{1}{4} \quad \text{Var}\left(\frac{1}{4n} \sum \frac{(Y_i - \mu)^2}{\sigma^2}\right) = \frac{1}{8n}$$

$$E\left[\frac{1}{4n} \sum (Y_i - \mu)^2\right] = \frac{1}{2} \sigma^2 \quad \text{Var}\left(\frac{1}{4n} \sum (Y_i - \mu)^2\right) = \frac{\sigma^4}{2n}$$

$$\text{So } E[\hat{\sigma}^2] = E\left[\frac{1}{2n} \sum (X_i - \mu)^2\right] + E\left[\frac{1}{4n} \sum (Y_i - \mu)^2\right] = \frac{1}{2} \sigma^2 + \frac{1}{2} \sigma^2 = \boxed{\sigma^2}$$

$$\text{Var}(\hat{\sigma}^2) = \text{Var}\left(\frac{1}{2n} \sum (X_i - \mu)^2\right) + \text{Var}\left(\frac{1}{4n} \sum (Y_i - \mu)^2\right) = \frac{\sigma^4}{2n} + \frac{\sigma^4}{2n} = \boxed{\frac{\sigma^4}{n}}$$

- (b) Since $E[\hat{\sigma}^2] = \sigma^2$, it is unbiased, and since $\lim_{n \rightarrow \infty} \text{Var}(\hat{\sigma}^2) = \lim_{n \rightarrow \infty} \frac{\sigma^4}{n} = 0$, it is also consistent.

2. (20 points) Random variables $Y_1, Y_2, \dots, Y_n$ are iid with pdf $f(y) = \frac{1+\theta}{\theta} y^{1/\theta}$ for $0 < y < 1$ and $\theta > 0$.

Find the MLE of θ . (Note: You don't have to check the second derivative for this problem.)

$$L(\theta) = \prod_{i=1}^n \frac{1+\theta}{\theta} y_i^{1/\theta} = \left(\frac{1+\theta}{\theta}\right)^n \prod_{i=1}^n y_i^{1/\theta}$$

$$\ell(\theta) = n \log\left(\frac{1+\theta}{\theta}\right) + \sum_{i=1}^n \frac{1}{\theta} \log y_i = n \log(\theta+1) - n \log \theta + \frac{1}{\theta} \sum \log y_i$$

$$\frac{d}{d\theta} \ell(\theta) = \frac{n}{\theta+1} - \frac{n}{\theta} - \frac{1}{\theta^2} \sum \log y_i = n \left(\frac{\cancel{\theta} - (\cancel{\theta} - 1)}{\theta(\theta+1)} \right) - \frac{1}{\theta^2} \sum \log y_i \stackrel{\text{set}}{=} 0$$

$$-\frac{n}{\theta+1} = \frac{1}{\theta} \sum \log y_i$$

$$-n = \sum \log y_i + \frac{1}{\theta} \sum \log y_i$$

$$\frac{1}{\theta} \sum \log y_i = -n - \sum \log y_i$$

$$\theta = - \frac{\sum \log y_i}{n + \sum \log y_i}$$

Ordinarily, we would check to make sure that $\frac{d^2}{d\theta^2} \ell\left(- \frac{\sum \log y_i}{n + \sum \log y_i}\right) < 0$ but we are skipping that step, so our MLE is

$$\hat{\theta} = - \frac{\sum \log Y_i}{n + \sum \log Y_i}$$

3. (20 points) Random variables $X_1, X_2, \dots, X_m$ are iid $N(\theta, 1)$ and, independently of the X 's, $Y_1, Y_2, \dots, Y_n$ are iid $N(\theta + 1, 1)$. Find a pivotal quantity based on $\bar{X} + \bar{Y}$ and use it to construct a 95% confidence interval for θ .

$$\bar{X} \sim N(\theta, \frac{1}{m}) \quad \bar{Y} \sim N(\theta+1, \frac{1}{n})$$

$$\bar{X} + \bar{Y} \sim N(\theta + \theta + 1, \frac{1}{m} + \frac{1}{n})$$

$$\sim N(2\theta + 1, \frac{1}{m} + \frac{1}{n})$$

$$\frac{\bar{X} + \bar{Y} - (2\theta + 1)}{\sqrt{\frac{1}{m} + \frac{1}{n}}} \sim N(0, 1)$$

This is a P.Q. because it depends only on the statistic $(\bar{X} + \bar{Y})$ and the parameter θ , and its dist. does not depend on θ .

$$\text{So } P(-1.96 < \frac{\bar{X} + \bar{Y} - (2\theta + 1)}{\sqrt{\frac{1}{m} + \frac{1}{n}}} < 1.96) = 0.95$$

$$P(-1.96 \sqrt{\frac{1}{m} + \frac{1}{n}} < \bar{X} + \bar{Y} - (2\theta + 1) < 1.96 \sqrt{\frac{1}{m} + \frac{1}{n}}) = 0.95$$

$$P(-1.96 \sqrt{\frac{1}{m} + \frac{1}{n}} - (\bar{X} + \bar{Y}) < -(2\theta + 1) < 1.96 \sqrt{\frac{1}{m} + \frac{1}{n}} - (\bar{X} + \bar{Y})) = 0.95$$

$$P(-1.96 \sqrt{\frac{1}{m} + \frac{1}{n}} - (\bar{X} + \bar{Y}) + 1 < -2\theta < 1.96 \sqrt{\frac{1}{m} + \frac{1}{n}} - (\bar{X} + \bar{Y}) + 1) = 0.95$$

$$P(\frac{1.96}{2} \sqrt{\frac{1}{m} + \frac{1}{n}} + \frac{\bar{X} + \bar{Y}}{2} - \frac{1}{2} > \theta > -\frac{1.96}{2} \sqrt{\frac{1}{m} + \frac{1}{n}} + \frac{\bar{X} + \bar{Y}}{2} - \frac{1}{2}) = 0.95$$

So the 95% c.i. for θ is

$$\frac{\bar{X} + \bar{Y}}{2} - \frac{1}{2} \pm \frac{1.96}{2} \sqrt{\frac{1}{m} + \frac{1}{n}}$$

because we multiply through by a negative number, must reverse the inequality

$$\begin{pmatrix} -5 < -4 \\ 5 > 4 \end{pmatrix}$$

4. (20 points) A random variable $Y \sim U(3, 3\theta)$ and it is of interest to test $H_0: \theta = 4$ vs. $H_a: \theta < 4$.
- The test has the form: Reject H_0 if $Y < c$. Find c such that the test has Type I error rate 0.05.
 - Calculate an expression (in terms of θ) for the power of the test in (a). What is the power when $\theta = 3$?

(a) Assuming H_0 is true, $Y \sim U(3, 12)$ so choose c so that

$$P(Y < c) = 0.05 \text{ if } H_0 \text{ is true}$$

$$P(Y < c) = \int_3^c \frac{1}{9} dy = \frac{1}{9} y \Big|_3^c = \frac{1}{9} (c-3) \quad \text{for } 3 < c < 12$$

Set this equal to 0.05 and solve for c

$$\frac{1}{9} (c-3) = 0.05$$

$$c-3 = 9(0.05)$$

$$c = 3 + 9(0.05) = \boxed{3.45}$$

(b) Power = $P(Y < 3.45)$ when $Y \sim U(3, 3\theta)$

$$P(Y < 3.45) = \int_3^{3.45} \frac{1}{3\theta-3} dy = \frac{1}{3(\theta-1)} y \Big|_3^{3.45} = \frac{1}{3(\theta-1)} (3.45-3)$$

$$= \frac{0.45}{3(\theta-1)}$$

$$\text{If } \theta=3 \text{ this is } \frac{0.45}{3(2)} = \boxed{0.075}$$

5. (20 points) An estimator of $\mu_1 - \mu_2$ is to be based on $\bar{X} - \bar{Y}$, where $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ and $\bar{Y} = \frac{1}{2n} \sum_{i=1}^{2n} Y_i$ (there are two times as many Y 's as X 's). The X 's are independent with mean μ_1 and variance 9, and the Y 's are independent with mean μ_2 and variance 9. (And the X 's are independent of the Y 's.)
- Provide an expression for an approximate 90% confidence interval for $\mu_1 - \mu_2$ based on these data.
 - How large does n need to be for the width of the interval (upper limit minus lower limit) to be no more than 3?

$$\begin{aligned} (a) \quad E[X_i] &= \mu_1, \quad \text{Var}(X_i) = 9 & E[Y_i] &= \mu_2, \quad \text{Var}(Y_i) = 9 \\ E[\bar{X}] &= \mu_1, \quad \text{Var}(\bar{X}) = \frac{9}{n} & E[\bar{Y}] &= \mu_2, \quad \text{Var}(\bar{Y}) = \frac{9}{2n} \\ E[\bar{X} - \bar{Y}] &= \mu_1 - \mu_2 & \text{Var}(\bar{X} - \bar{Y}) &= \frac{9}{n} + \frac{9}{2n} = \frac{27}{2n} \end{aligned}$$

If n is large then $\bar{X} - \bar{Y} \sim N(\mu_1 - \mu_2, \frac{27}{2n})$
 So an approximate 95% c.i. is $\bar{X} - \bar{Y} \pm 1.645 \sqrt{\frac{27}{2n}}$

(b) width of the interval is $2(1.645) \sqrt{\frac{27}{2n}} \leq 3$

$$\frac{2}{3} (1.645) \sqrt{27} \leq \sqrt{2n}$$

$$\frac{4}{9} (1.645)^2 27 \leq 2n$$

so we need $n \geq \frac{6 (1.645)^2}{1} = 16.236$

or $n \geq 17$ if we are going to consider only integers